

AN EMPIRICAL STUDY OF THE WILCOXON-MANN-WHITNEY U TEST FOR SHIFT IN AR(1) DEPENDENT VARIABLES

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Key words: Asymptotic power; dependence between samples; dependence within samples; normal approximation; shift; Wilcoxon-Mann-Whitney test.

Summary: The Wilcoxon-Mann-Whitney test is used to test for equality of medians for two populations. The purpose of this study is to assess the effect of relaxing the assumption of independence of the variables on the distribution and its power function. When using this test to test for shift in an AR(1) process at a known point in time the two samples (prior to and after the alleged point of shift) exhibit both dependence within and between samples. This type of dependence is compared with dependence within the samples only. It will be shown that the presence of dependence (as opposed to independence) has a considerable effect on the appearance of the distribution as well as on the power function. Cases where the sample sizes are equal and not equal are considered.

1. Introduction

The Wilcoxon-Mann-Whitney U test is a well-known nonparametric test for equality of medians of two populations. The usual assumption made is that the two samples are independent. However, there are some situations where this assumption of independence is violated. Examples of such situations are

1. Comparing the effects of treatments 1 and 2. Each treatment is given to n randomly selected subjects. Due to a shortage of subjects some subjects

AMS: 62G30, 00A72, 62M10

are used twice in the experiment. Since some subjects are used twice, the two samples cannot be regarded as independent. This experiment introduces dependence between the samples. Within each sample the observations can be regarded as independent.

2. Some variable is observed over time (time period 1). After a period of time had elapsed the variable is again observed over time (time period 2). The interest is in whether there has been a shift in the median of the variable from time period 1 to time period 2. If the two time periods are sufficiently far apart, the two samples can be regarded as independent of each other and only dependence within samples will be present.

The Wilcoxon-Mann-Whitney U test under the assumption of dependence between samples (example 1) was studied by Govindarajulu (1975) and that under the assumption of dependence within samples (example 2) by Serfling (1968). Hollander, Pledger and Lin (1974) looked at the test under an assumption similar to that of Govindarajulu. It was assumed that some of the observations in the two samples could be paired, while others were independent. Pettitt and Siskind (1981) did a study similar to that of Hollander, Pledger and Lin, but made some assumptions about the form of the distributions from which the samples were drawn. Yue and Wang (2002) did a simulation study to assess the influence of AR(1) correlation on the Wilcoxon-Mann-Whitney U test for detecting a shift in median by comparing the type I error probabilities under dependence and independence. In each of the first 3 articles mentioned a Central Limit Theorem for the normalised Wilcoxon-Mann-Whitney U statistic was proved.

In this study dependence within and between samples will be considered. The U-distribution for this type of dependence will be compared with that where only dependence within samples is present. For the following reasons the focus of the study will be on the large sample case.

1. In the case of independent samples the normal approximation to the U test is generally used. The condition for using this test is that the minimum of the two sample sizes should be at least 4. A question of practical interest would be whether a similar approximation could be used for dependent samples and (if so) for what sample size this approximation could be used. It will be found that in such a case a considerably larger sample size (than 4) would be needed than for independence. In such an application the Central Limit Theorem results (large sample case) would be of interest.
2. An important application of the U test for dependent variables is to test for a shift in the median of a time series. In order to be able to monitor the trend of the series and identify such a point of shift, a reasonably long time series is needed. This results in the large sample case being of practical importance.

2. Definitions

Consider a time series of length $N = m + n$. A test for the presence of a shift of size c in the series after time m is to be performed. The two portions of the series will be denoted by X_i , $i = 1, 2, \dots, m$ and $Y_j = X_j + c$, $j = m + 1, m + 2, \dots, m + n$. It will be assumed that under the hypothesis of no shift ($c = 0$) the series X_i , $i = 1, 2, \dots, m + n$ is generated according to the AR(1) scheme

$$X_i = \mu + \phi(X_{i-1} - \mu) + a_i, \quad i = 2, 3, \dots, m + n$$

where $a_i \sim IN(0, 1)$, μ is the mean of the series and

$$X_1 - \mu = a_1 / (1 - \phi^2).$$

A test for shift can be based on the Wilcoxon-Mann-Whitney statistic

$$U_{m,n} = \sum_{i=1}^m \sum_{j=m+1}^{m+n} s(Y_j - X_i),$$

where

$$\begin{aligned} s(x) &= 1 \text{ if } X_i > Y_j \\ &= 0 \text{ otherwise.} \end{aligned}$$

3. Simulation results for equal sample sizes under the hypothesis of no shift – dependence between and within samples

First the distribution of U for the case $m = n$ and $c = 0$ will be approximated. In order to do this, 10 000 samples of size $N = 2n$ will be generated for various values of ϕ . The random variable U can assume any of the values $0, 1, \dots, n^2$.

3.1 Small sample sizes

Graphs for $N = 12$ and $\phi = -0.8, 0$ (independence) and 0.8 are shown next.

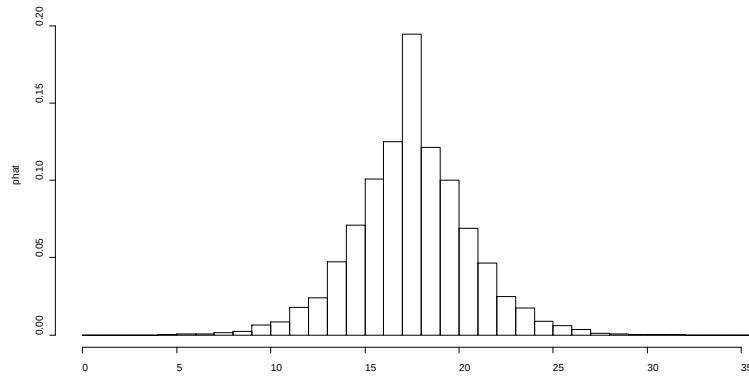


Figure 1(a). Simulated U -distribution for $N = 12$, $\phi = -0.8$

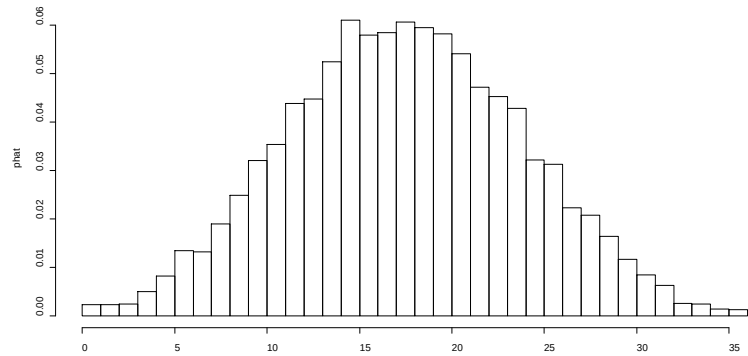


Figure 1(b). Simulated U-distribution for $N = 12$, $\phi = 0$

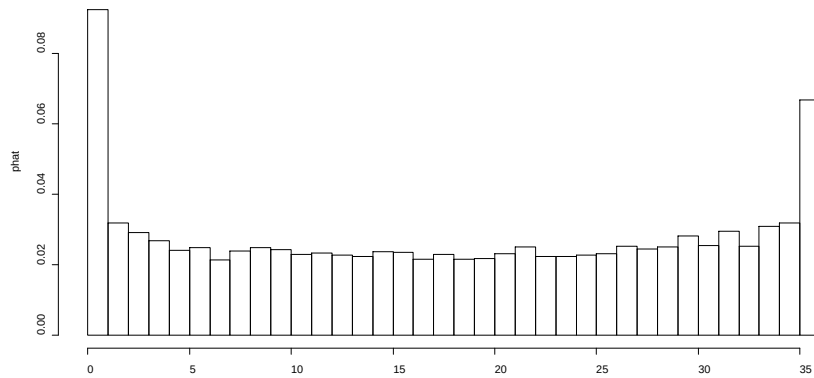


Figure 1(c). Simulated U-distribution for $N = 12$, $\phi = 0.8$

1. All distributions appear to be symmetrical.
2. For $\phi = -0.8$ (strong negative correlation) the values of u are more concentrated towards the middle of the scale. The graph is unimodal with maximum probability at $u = 18$.
3. For $\phi = 0$ (independence) the main concentration of values is towards the middle of the scale, but the values are more spread out than for negative correlation.

4. For $\phi = 0.8$ (strong positive correlation), there has been a shift of values from the middle of the scale towards the two ends with maximum probabilities at the two extremes.
5. The means for $\phi = -0.8$, 0 and 0.8 are 17.9895, 17.9704 and 18.1636 respectively. For all the distributions, the mean is approximately 18 i.e. halfway between the minimum and maximum values.
6. The variances for $\phi = -0.8$, 0, and 0.8 are 9.00279, 39.62292 and 139.0144 respectively. The increase in the variation of the u values as ϕ increases, as noted in 2, 3 and 4 above, is reflected in the increase of the variance as ϕ increases.

3.2 Medium size samples

Graphs for $N = 30$, $\phi = -0.8$, 0 (independence) and 0.8 are shown next.

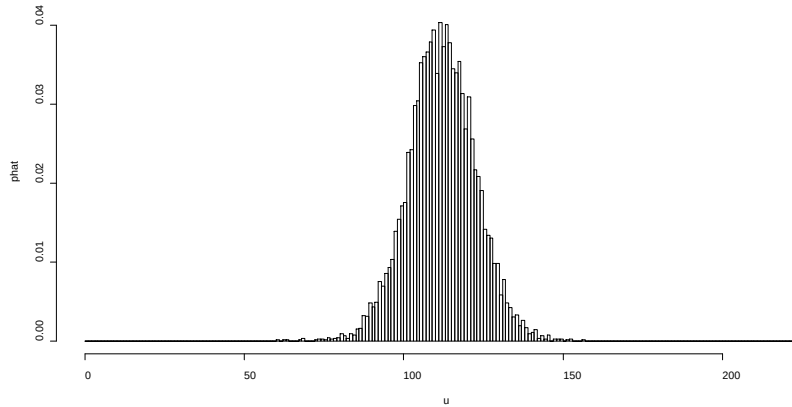


Figure 2(a). Simulated U distribution for $N = 30$, $\phi = -0.8$

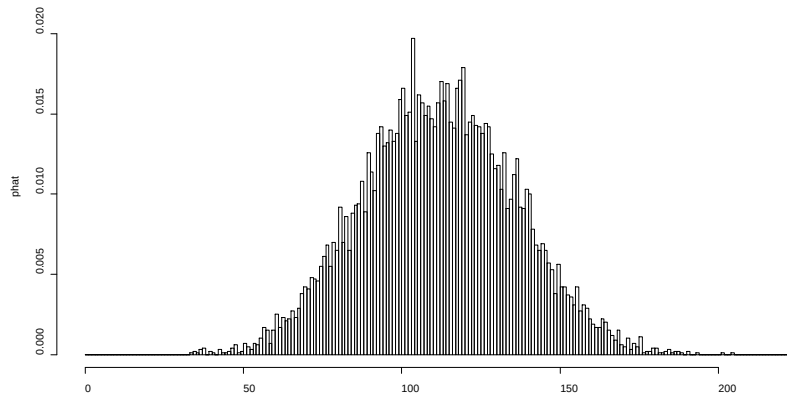


Figure 2(b). Simulated U distribution for $N = 30$, $\phi = 0$

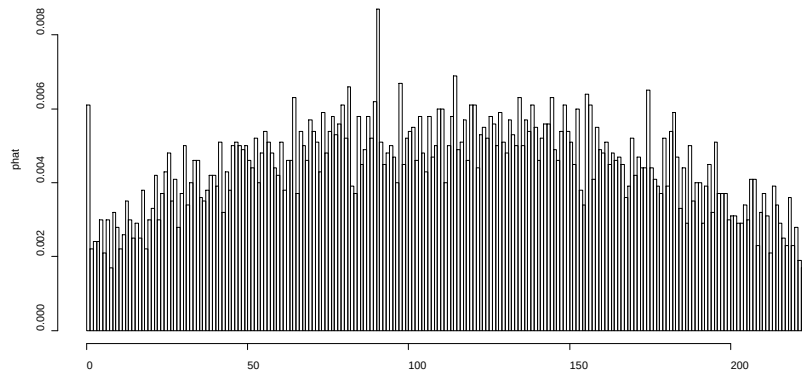


Figure 2(c). Simulated U distribution for $N = 30$ and $\phi = 0.8$

1. For $\phi = -0.8$ (strong negative correlation) the values of u are even more concentrated towards the middle of the scale than for $N = 12$. The distribution is more peaked in the centre part of the graph.

2. For $\phi = 0$ there is also a drift of values towards the middle part of the scale. The graph has a distinct symmetric, bell shaped appearance. This suggests that the normal distribution could approximate this distribution.
3. For $\phi = 0.8$ the dent that appeared in the graph for $N = 12$ starts disappearing. The graph is starting to show a peak in the middle of the scale. The sizes of the frequencies at the extremes of the scale are not as prominent.

3.3 Large size samples

Graphs for $N = 90$, $\phi = -0.8$, 0 (independence) and 0.8 are shown next. The u values were sorted into classes of length 15. The graph for $N = 240$ for $\phi = 0.8$ with classes of length 120 follows these graphs.

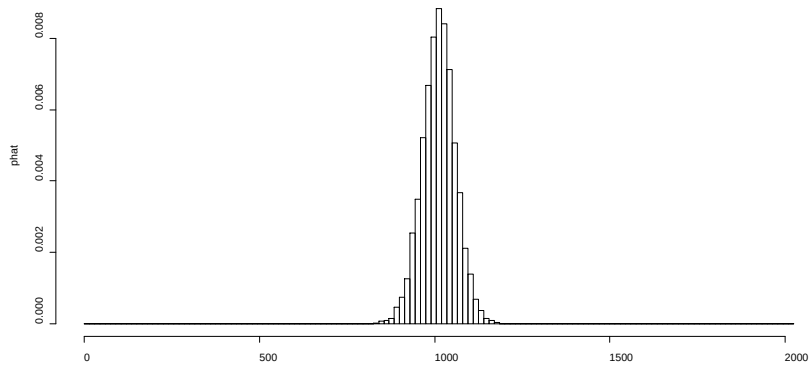


Figure 3(a). Simulated U distribution for $N = 90$, $\phi = -0.8$

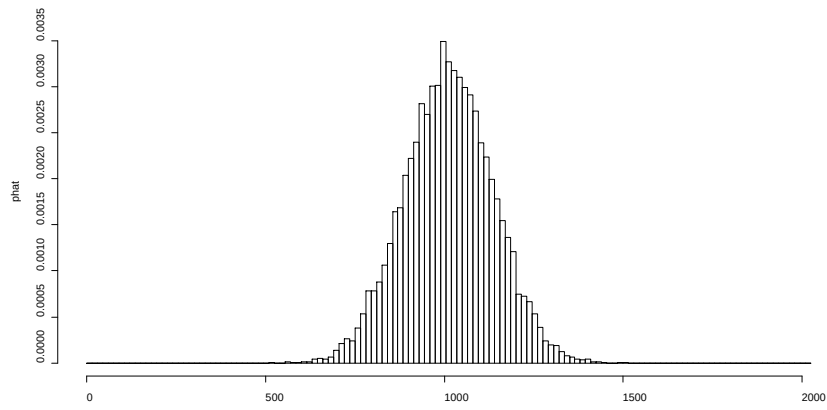


Figure 3(b). Simulated U distribution for $N = 90$, $\phi = 0$

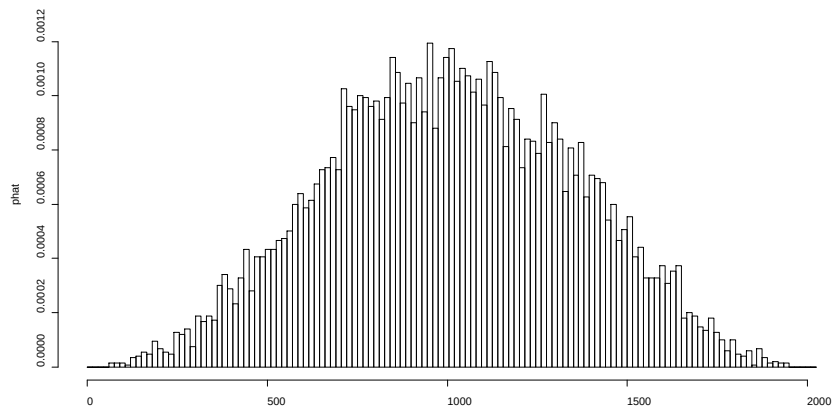


Figure 3(c). Simulated U distribution for $N = 90$, $\phi = 0.8$

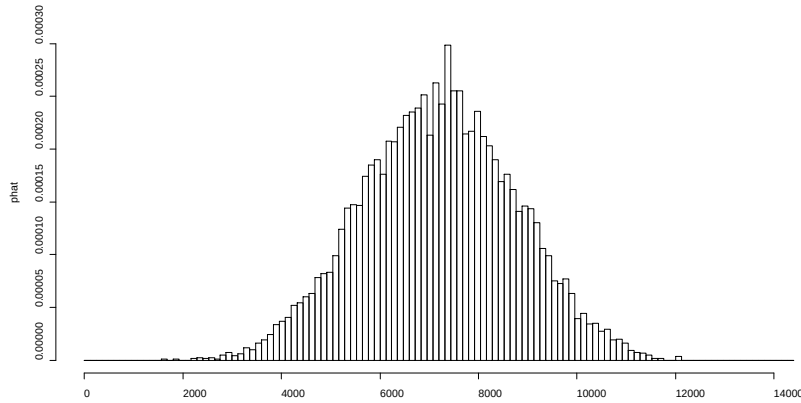


Figure 4. Simulated U distribution for $N = 240$, $\phi = 0.8$

1. All the above histograms have a symmetric, bell shaped appearance.
2. For $\phi = 0.8$ (strong positive correlation) a considerably larger sample size (than for independence or negative correlation) is needed to achieve the symmetric, bell shaped appearance of the graph.

3.4 Explanation for patterns observed in results

Each observation that occurs prior to the point of shift will be labelled an x value and each one after the point of shift a y value. Each value of U is based on an ordered sequence (in decreasing order of magnitude) of x and y values. Values of U at the extremes of the scale result from separated sequences of x 's and y 's e.g. for the case $N = 8$ the sequence $yyyyxxxx$ will yield $U = 0$ and $xxxxyyyy$ $U = 16$. Values in the middle of the scale will result from mixed sequences of x 's and y 's e.g. for the case $N = 8$ the sequence $yxyxyyx$ will yield $U = 10$. Under independence all the sequences of x 's and y 's are equally likely. Under positive dependence separated sequences are more likely and mixed sequences less likely than mixed ones when compared to independence. Under negative dependence the opposite is true.

The probability of each U value is the sum of the probabilities of all the sequences of x 's and y 's that result in this value. In the case of independence the probability that $U = u$ is $\frac{\text{number of sequences where } x\text{'s} > y\text{'s } u \text{ times}}{m+n C_m}$
i.e. each sequence has probability $1/(m+n C_m)$.

In the case of positive dependence smaller probabilities (than for independence) are added in the middle of the scale and larger ones (than for independence) at the extremes of the scale. In the case of negative dependence larger probabilities (than for independence) are added in the middle of the scale and smaller ones (than for independence) at the extremes of the scale. This explains the dent in the middle of the scale and the peaks at the two ends of the scale for positive dependence for $N = 12$ and the peaked appearance in the middle of the scale for negative dependence for $N = 12$.

As the overall sample size increases the number of mixed appearances increase at a much faster rate than the number of separated ones e.g. for $N = 8$ the ratio $\frac{\text{number of } N \text{ middle sequences}^{1})}{\text{number of } N \text{ extreme sequences}^{2})}$ is $\frac{51}{14} = 3.64$. For $N = 12$ this ratio increases to $\frac{599}{38} = 15.76^{3)}$.

This means that as N increases more and more probabilities are added in at U -values in the middle of the scale, while the probabilities added in at values at the extremes increase very little. This has the effect of removing the dent in the middle of the scale for positive dependence and increasing the peak in the middle for negative dependence as N increases.

1. For $N = 8$ the number of middle sequences is calculated from those resulting in values of u from 4–11. For $N = 12$ the number of middle sequences is obtained from those resulting in values of u from 12–23.
2. For $N = 8$ the number of extreme sequences is calculated from those resulting in values of u from 0–3 or 13–16. For $N = 12$ the number of extreme sequences is obtained from those resulting in values of u from 0–5 or 31–36.

3. The calculations were done by using the formulae used by Di Bucchianico (1999) and Wilcoxon, Katti and Wilcox (1973).

4. Simulation results for equal sample sizes under the hypothesis of no shift – dependence within samples

Suppose that a number of observations in the middle of the series are removed so that the two portions are independent. This results in only dependence within the samples being present. Simulation experiments indicate that the U-distribution in such a case has shapes that are similar than those shown in figures 1 to 4 with the following differences.

1. For negative ϕ the U-distribution for dependence within samples is more peaked than that for dependence within and between samples (has smaller variance).
2. For positive ϕ the U-distribution for dependence within samples is more spread out than that for dependence within and between samples (has greater variance).

These differences are due to the effect of the adjacent values (those close to where the x and y values meet) of the two portions of the series when considering sequences with dependence between and within samples. For sequences with dependence within and between samples the adjacent x and y values are mixed to a greater extent (than for dependence within samples) for positive ϕ and separated to a greater extent (than for dependence within samples) for negative ϕ .

The ratios of the standard deviations for the two types of dependence are shown in the table below.

Table 1. Ratio of the standard deviation (U |Dependence within and between samples) to standard deviation (U |Dependence within samples) for various values of N and ϕ .

N/ϕ	-0.8	-0.4	0.4	0.8
12	1.1464	1.0274	0.9603	0.8655
60	1.0762	1.0107	0.98	0.9152
90	1.0491	1.0038	0.9886	0.9618
120	1.0255	1.0096	0.9979	0.9796
240	1.0133	0.9892	0.9858	0.9871
360	1.0042	0.9929	0.9926	0.9910

The results in the above table suggest that for negative ϕ this ratio will converge to 1 from above and for positive ϕ it will converge to 1 from below as N increases. The tests for the two dependence situations (within and between samples and only within samples) appear to be asymptotically equivalent. The value of N for which this holds will depend on the value of the dependence parameter ϕ . The further $|\phi|$ is from 0, the larger the value of N needed to achieve this.

5. Practical issues concerning the U test for dependent variables

5.1 Sample size needed to achieve normality

The patterns shown in the graphs in the previous section suggest that for a sufficiently large sample size the normal distribution can approximate the U-distribution fairly well. The next table is a summary of chi-square goodness-of-fit tests (observed versus expected normal distribution frequencies) that were performed for values of $N = 12, 30, 60, 120$ and 240 for $\phi = -0.8, -0.6, \dots, 0, 0.2, \dots, 0.8$. All the tests are based on 10 000 simulated u values. Since the numbers of classes involved in the different tests differ, the chi-square values cannot be compared directly. For the case $N = 240$ the chi-

square values for $\phi = -0.6, -0.4, \dots, 0.4$ are omitted. Based on the results for smaller values of N , the normal approximations to the U-distribution for these cases are obviously sufficiently close.

Table 2. The chi-square goodness-of-fit test statistic for the normal approximation to the U-distribution for various values of N and ϕ .

ϕ/N	12	30	60	120	240
-0.8	456.81***	32.13***	35.61***	27.01***	8.09
-0.6	61.32***	17.40***	23.01***	16.4	
-0.4	63.96***	18.05**	6.16	11.32	
-0.2	60.84***	11.80	7.47	16.25	
0	92.15***	14.76	10.23	27.42	
0.2	130.84***	30.32***	18.06	31.57	
0.4	319.53***	84.67***	26.90	35.95	
0.6	772.57***	161.65***	67.61***	52.37**	41.53
0.8	1807.742***	476.02***	291.87***	124.46***	80.59

* Significant at the 10% level of significance.

** Significant at the 5% level of significance.

*** Significant at the 1% level of significance.

1. For $N = 12$ the fit is based on the classes 0, 1,..., 36.
2. For $N = 30$ the fit is based on the classes 0–15, 16–30,..., 211–225.
3. For $N = 60$ the fit is based on the classes 0–30, 31–60,..., 871–900.
4. For $N = 120$ the fit is based on the classes 0–60, 61–120,..., 3541–3600.
5. For $N = 240$ the fit is based on the classes 0–120, 121–240,..., 14281–14400.
6. For cases where expected frequencies are small the classes were combined.
7. The normal approximations are based on a mean of $\frac{n^2}{2}$ and variance estimated from the simulated sample.

For $N = 60$ the normal approximation appears to be valid for weak correlation i.e. $|\phi| \leq 0.4$. For $N = 120$ the normal approximation appears to be valid for moderate negative correlation i.e. $-0.6 \leq \phi < 0$ and weak positive correlation i.e. $0 < \phi \leq 0.4$. For $N = 240$ the normal approximation

appears to be valid for $|\phi| \leq 0.8$. The results in the above table for $\phi = 0$ (independence) seems to cast doubt about using the normal approximation for a value of N as low as 8 i.e. $n = 4$.

5.2 Dependence within and between samples for $m \neq n$

Only the cases $m = \frac{n}{2}$ and $n = \frac{m}{2}$ were considered. The U-distribution was simulated for various values of $N = m + n$ and for $\phi = -0.8, -0.6, \dots, 0, 0.2, \dots, 0.8$. The simulation results confirmed the result proved by Serfling (1968) i.e. asymptotic normality of the U-distribution. Chi-square goodness-of-fit tests suggest that for these cases larger values of N are needed to achieve approximate normality than for the case $m = n$.

5.3 Formulae for the mean and variance of U

5.3.1 Mean

The mean of U for dependence within and between samples

The simulation results suggest that

$$\text{mean (U)} = \frac{mn}{2} \text{ for all values of } \phi.$$

This result will follow if it can be shown that the U-distribution is symmetrical. The symmetry follows from the following arguments.

Under H_0 the two portions will be labelled $X_i, i = 1, 2, \dots, m$ and $X_j, j = m + 1, m + 2, \dots, m + n$. The value of $U_{m,n} = u$ arises from an ordered sequence of x_i 's and x_j 's (in increasing order of magnitude), where the x_j 's are less than the x_i 's u times. Then the sum of the ranks of the x_i 's in the combined sequences of x_i 's and x_j 's is $u + \frac{m(m+1)}{2}$. The sequence on which these calculations are based will be referred to as sequence 1.

Consider an altered sequence of x_i 's and x_j 's where each x_i and x_j in sequence 1 are replaced by $x_{m+n+1-i}$ and $x_{m+n+1-j}$ respectively. This sequence will be referred to as sequence 2. For this sequence only the indices

are changed (not the actual values). Therefore the sum of the ranks of the x_i 's in sequence 2 will be

$$\begin{aligned} S &= \text{Total sum of ranks} - \text{sum of ranks of the } x_j\text{'s} \\ &= \frac{(m+n)(m+n+1)}{2} - \left[u + \frac{n(n+1)}{2} \right] \\ &= mn - u + \frac{m(m+1)}{2}. \end{aligned}$$

It follows that the value of $U_{m,n}$ for this sequence is $mn - u$.

If the ordered sequence 1 is written as

$$X_{i(1)} < X_{i(2)} < \dots < X_{i(m+n-1)} < X_{i(m+n)},$$

the ordered sequence 2 will be written as

$$X_{mn-i(1)+1} < X_{mn-i(2)+1} < \dots < X_{mn-i(m+n-1)+1} < X_{mn-1(m+n)+1}.$$

When calculating the probability of sequences 1 and 2, the multivariate normal distribution with mean vector $\underline{\mu} = (\mu\mu\dots\mu)'$ and covariance matrix Σ with the (i, j) -th entry $\frac{\phi^{|i-j|}}{1-\phi^2}$ will have to be integrated. The region of integration for sequence 1 is

$$\begin{aligned} -\infty &< x_{i(1)} \leq x_{i(2)}, -\infty < x_{i(2)} \leq x_{i(3)}, \dots, -\infty \\ &< x_{i(m+n-1)} \leq x_{i(m+n)}, -\infty < x_{i(m+n)} < \infty \end{aligned}$$

and that for sequence 2

$$\begin{aligned} -\infty &< x_{mn-i(1)+1} \leq x_{mn-i(2)+1}, \dots, -\infty < x_{mn-i(m+n-1)+1} \\ &\leq x_{mn-i(m+n)+1}, -\infty < x_{mn-i(m+n)+1} < \infty. \end{aligned}$$

Sequence 1 and its associated region of integration can be viewed as being generated by the backward form of the AR(1) process i.e.

$$X_i = \mu + \phi(X_{i-1} - \mu) + a_i, \quad i = 2, 3, \dots, m+n.$$

Sequence 2 and its associated region of integration can be viewed as being generated by the corresponding forward form (Box and Jenkins, 1970) of the AR(1) process i.e.

$$X_i = \mu + \phi(X_{i+1} - \mu) + b_i, \quad i = 1, 2, \dots, m+n-1 \text{ where } b_i \sim IN(0, 1).$$

Hence the two sequences are generated by the same mechanism with sequence 2 the back to front version (in time) of sequence 1. Since the orders in which the component variables in sequences 1 and 2 appear are the same and the integrations involve the same integration regions and density functions being integrated, the probabilities of the two sequences are the same.

If this argument is repeated for all the sequences of x_i 's and x_j 's that result in $U_{mn} = u$ and $U_{mn} = mn - u$, then it follows that

$$P(U_{mn} = u) = P(U_{mn} = mn - u)$$

and therefore

$$E(U_{m,n}) = \frac{mn}{2}.$$

5.3.2 Large sample variance for $m = n$

The variance for $N = 30, 60, 90, 120, 180, 240$ for $\phi = -0.8, -0.6, \dots, 0.8$ was estimated from simulation results. For sufficiently large N the variance was found to be quite close to $var(U|\phi) = \frac{\rho(\phi)n^2(2n+1)}{12}$, where $\rho(\phi) = \frac{1 + (6/\pi) \arcsin(\phi/2)}{1 - (6/\pi) \arcsin(\phi/2)}$, which is the formula obtained when using the Pearson grade correlation result for the bivariate normal distribution in the formula (4.2) given by Serfling. The ratios for the simulation standard deviations to the theoretical ones are given in the table below.

Table 3. Ratios of simulation standard deviations of U for dependence within and between samples and theoretical standard deviations ($m = n$)

N/ϕ	-0.8	-0.6	-0.4	-0.2	0.2	0.4	0.6	0.8
30	1.06889	1.027414	1.014296	1.000482	0.989415	0.989554	0.966465	0.924059
60	1.049765	1.015127	1.005637	1.010623	0.99816	0.991332	0.991519	0.96048
90	1.019157	1.012196	1.006406	1.003341	0.996874	0.994988	0.986456	0.966357
120	1.022294	1.008767	1.000839	1.002728	0.997786	1.002851	0.990829	0.980051
180	1.018393	0.997518	1.005121	1.000759	0.996384	0.996562	0.996202	0.985052
240	1.010747	1.00738	0.996058	1.000019	1.003033	0.983045	1.009398	0.989496

As can be expected, the weaker the dependence the faster the convergence to 1.

5.3.3 Large sample variance for $m \neq n$

Only the cases $n = \frac{m}{2}$ and $m = \frac{n}{2}$ were considered. Simulation results suggest that their variances are the same. As for the case $m = n$ the variances calculated by using simulation is a good approximation for the theoretical result $var(U|\phi) = \frac{\rho(\phi)mn(m+n+1)}{12}$. The value of $N = m + n$ for which this approximation is reasonable is considerably larger than for the case $m = n$. The ratios of the simulated and theoretical standard deviations for the case $N = 540$, $m = \frac{n}{2}$ are shown in the table below.

Table 4. Ratio of simulated and theoretical standard deviations where $N = 540$, $m = \frac{n}{2}$ for various values of ϕ .

ϕ	-0.8	-0.6	-0.4	-0.2	0.2	0.4	0.6	0.8
ratio	1.020109	1.0136	1.002567	1.005271	0.998756	0.996445	1.002878	1.005285821

6. Large sample power of the U-test for dependence within and between samples

Yue and Wang (2002) presented graphs showing power calculations based on the normal approximation to the U-distribution for the case $m = n = \frac{N}{2}$ for $N = 20, 40, \dots, 120$ and $\phi = -0.8, -0.6, \dots, 0.8$. Since the accuracy of the normal approximation is doubtful for some of these combinations of (N, ϕ) values (see section 5.1), some these graphs may be inaccurate. The graphs below are all based on large N (180 or greater). For these values of N there is less doubt about the accuracy of the normal approximation. The graphs are based on simulating 10 000 values of the U-distribution for $m = n = \frac{N}{2}$ with a shift of size c after the m -th observation and calculating the proportion of values less than $\frac{n^2}{2} - z_a \left[\frac{\rho(\phi) n^2 (2n + 1)}{12} \right]^{\frac{1}{2}}$ where z_a is such that an area a under the standard normal curve is above it.

The results of some power calculations for $N = 180, 240$ and 300 are shown in the graphs that follow.

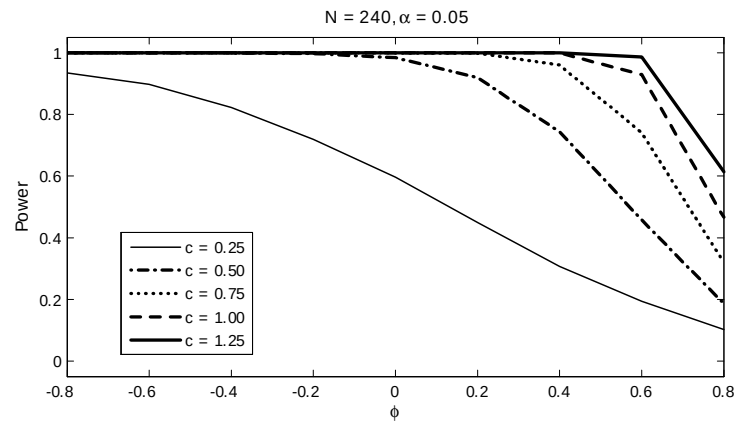


Figure 5(a). Large sample power of the U-test for various ϕ with $N = 180$ a shift of c and $\alpha = 0.05$.

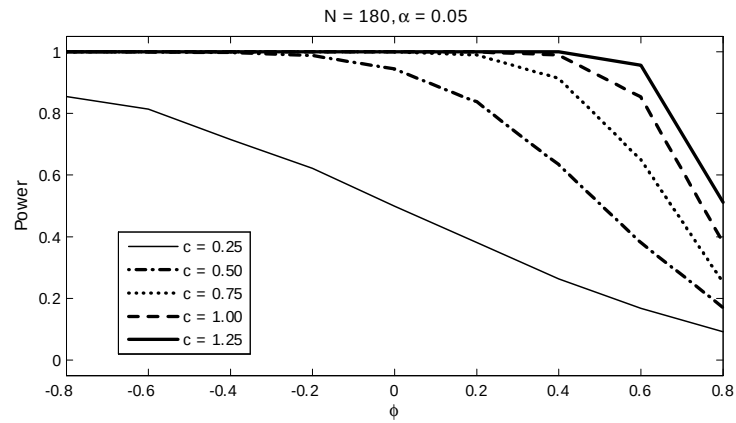


Figure 5(b). Large sample power of the U-test for various ϕ with $N = 240$, a shift of c and $\alpha = 0.05$.

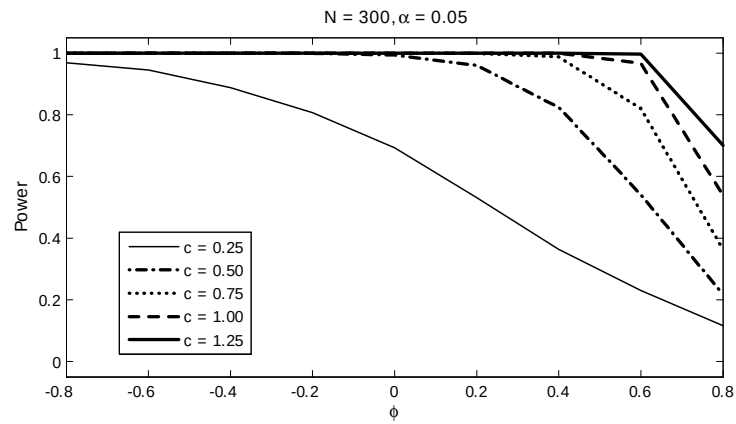


Figure 5(c). Large sample power of the U-test for various ϕ with $N = 300$, a shift of c and $\alpha = 0.05$.

1. For negative ϕ the power of the test is greater than that under independence. For positive ϕ the power of the test is less than that under independence.
2. The power increases as N increases for a fixed ϕ , α and c .
3. Even for moderate sample sizes the power for negative ϕ is high e.g. for $N = 60$, $\phi = -0.6$, $\alpha = 0.05$, and $c = 0.5$ the power is 0.9288.
4. Even for large sample sizes the power for high positive ϕ is low e.g. $N = 300$, $\phi = 0.8$ and $c = 0.5$ the power is 0.218.
5. The smaller the value of α , the less the power for fixed N , ϕ and c e.g. for $N = 240$, $\phi = -0.6$ and $c = 0.25$ the powers for $\alpha = 0.10, 0.05, 0.01$ are 0.9455, 0.8983 and 0.714 respectively.
6. The power is more sensitive to changes from medium to large positive correlation than to changes from medium to large negative correlation as can be seen from the next table.

The comments 1 and 2 above confirm comments on the power of the tests made in section 4 of the article by Yue and Wang (2002). Their simulation results were based on fewer simulated values (2000) and applied to smaller N values (120 or more).

Table 5. Medium to large negative and positive correlation for $N = 240$, $\alpha = 0.10$, $c = 0.25$.

ϕ	power
-0.6	0.9455
-0.8	0.9682
0.6	0.3163
0.8	0.1868

The change in power when ϕ changes from -0.8 to -0.6 is $0.9682 - 0.9455 = 0.0227$. The change when the value of ϕ changes from 0.6 to 0.8 is $0.3163 - 0.1868 = 0.1295$.

The changes in power (with respect to independence) for negative and positive ϕ can be explained in the following way. For negative ϕ the U-distribution is peaked i.e. the values are all fairly close to the center of the distribution. For positive ϕ the U-distribution is more spread out and the tails

of the distribution thicker. For negative correlation the probability of getting a value removed c units or more from the center of the U-distribution is smaller (and the power greater) than that for a value removed the same distance from the center for positive correlation.

7. Further studies

1. Identification of the point of shift.

In this study it was assumed that the possible point of shift in the series is known. There could be situations where there might be some doubt about the point of shift. The most likely point of shift could be identified by using maximum power as a selection criterion.

2. Estimation of the variance of U from the data.

Possible approaches to estimating of the variance of U from the data could include the following.

1. To replace ϕ by $\hat{\phi}$, the lag 1 autocorrelation of the series, in the formula
$$\text{var}(U|\phi) = \frac{\rho(\phi)mn(m+n+1)}{12}.$$
2. To estimate ϕ by $\hat{\phi}$ (as explained under 1) and to estimate the variance via simulation experiments for a given m , n and ϕ .
3. To use the consistent estimator of the variance of U suggested by Serfling (1968) [formula (5.2) of section 5].

These methods of estimating the variance of U could be compared on the basis of bias and mean square error.

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